

Nicolas Jao

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Education

McMaster University

Bachelor of Technology, Automation Systems Engineering Technology Co-op Program

Hamilton, ON

2022 - 2026

Mohawk College – Combined Degree-Diploma Program

Advanced Diploma in Chemical Engineering Technology, ACBSP Certificate of Business

Hamilton, ON

2022 - 2026

Experience

Sofina Foods

Engineering Co-op

Mississauga, ON

May 2026 - Aug 2026

- Led commissioning, qualification, and validation (CQV) for 4 capital projects at 4 plant sites
- Produced 2 job aid documents and collected signatures for a new sanitation water pump installation at the Orlando plant
- Developed 1 job aid, 1 CL, 1 CIL, 1 LOTO SOP for a new prosciutto vacuum packaging machine at the Goreway plant
- Created downtime reports and tracked root causes for Line 5 thermoformer and presented findings at the Humberline plant, aiding in the final decisions to solve the machine's issues
- Analyzed downtime, leaker rate, and machine utilization for Cryovac 3 and presented findings at the Burlington plant, giving the Process Engineer insights on how speed affects the machine's efficiency
- Designed an Ethernet infrastructure layout in **AutoCAD** for the Humberline plant for 12 Ethernet drops to existing HMIs outside salami dry rooms and prepared the Authorization for Expenditure (AFE), modernizing the plant's connectivity

Mecsmart Systems

Electrical Design Co-op

Newmarket, ON

Jan 2025 - Aug 2025

- Designed electrical systems for 10+ industrial manufacturing machines such as MIG welders, laser welders, riveters, and end-of-line testers for the automotive industry, creating an impact on the automotive supply chain
- Drafted 3+ electrical schematics and 5+ panel layouts for the electricians using **EPLAN Electric P8** and **SolidWorks**
- Improved electrical department efficiency by 30%, creating new standard procedures, implementing new **EPLAN** and **Excel** features and updating documentation for future employees
- Gained valuable experience reading 50+ data sheets, wiring diagrams, terminal pinouts, instruction manuals, user manuals, customer quotes and project management technical documents to complete daily tasks

Wiz Robotics

Robotics Engineering Co-op

Richmond Hill, ON

May 2023 - Aug 2023

- Programmed PID controllers on motors and designed odometry systems on FTC robots using **Java** to aid the research of two award-winning robotics teams, while providing mentorship and experience to the students
- Spearheaded the mechatronics skills development of 20+ students by instructing the fundamentals of the **Arduino** ecosystem, electrical circuits, and programming to build their own robot projects over 17 weeks

Projects

CAPSTONE Design Project

Sushi Assembly Line

Jan 2026 - Dec 2026

- Designed a fully automated sushi production machine integrating material handling, process sequencing, rolling and slicing with minimal operator input, with objectives in safety, consistency, and food grade materials
- Led the mechanical design with **Onshape** and electrical and controls system design with an **Arduino** controller

SMRTTECH 4ID3 Final Project

Aegis Grid

Jan 2026 - Mar 2026

- Designed an IoT network to monitor the health of electrical equipment in the power grid for preventative maintenance
- Created a block diagram, chose components for the proposed industrial solution, and validated its value proposition
- Assembled a proof-of-concept circuit and programmed the **Arduino** controllers and cloud network with **C++** and **Python**

Mac Falcons Robotics Team

Control Team Member

Sep 2023 - Apr 2026

- Applied control theory using **C/C++** and **MATLAB** to design the movement systems for the team's 4 robots
- Tested linear actuator motors and wrote code for controller board firmware with embedded systems **C/C++**
- Went to Shenzhen, China for the RoboMaster international competition with 100+ teams

Skills

3D Design: 3D Printing, SolidWorks, AutoCAD, Onshape, Blender, Unity, mills, bandsaws, power tools

Electrical: EPLAN Electric P8, KiCad, oscilloscopes, multimeters, soldering irons, crimpers

Programming: VS Code, Python, Java, C/C++, Arduino, Git

Other Software: Allen-Bradley RSLogix 5000, Siemens TIA Portal, FESTO DeviceCare SFE100, Fanuc RoboGuide, Honeywell UniSim Design R461, e2 Studio, NI LabVIEW, Intel Quartus II, MATLAB, Excel, Microsoft Office